General histology

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LG 3/ Blood tissue

List of Contents:

- Learning outcome.
- composition of blood .
- structure of different type of blood cells.

Learning outcome.

- Student have to know composition of blood (plasma & cells)
- Student should know different types of blood cells there function, normal range, shape and structure of each cell type.

Blood

- Is a specialized type of connective tissue constitute about 7 % of the body weight & the total blood volume in human is about (5-6) liters

  (depending on the body size) . It consists of
- 1-blood plasma.
- 2-Formed elements.

1- Blood Plasma

- Is an aqueous solution in which the formed elements are suspended and it consists of :

1- Water : make about 90% of the plasma .

2-Solutes: make about 10% of the plasma and consists of:

- a-Plasma proteins make (7%), include
- *Albumin (maintains the osmotic pressure of the blood).
- *Globulin (antibodies).
- *Fibrinogen forms the fibrin in final step of coagulation.

- b-Organic compounds (2.1%) such as amino acids, glucose, vitamins, hormones & lipoproteins.
- c-Inorganic salts (0.9%) such as sodium, potassium & calcium salts.

Blood Plasma

Water
90%

Solutes 10%

Plasma proteins
7%
*Albumin, Globulin (antibodies)
*Fibrinogen

Organic compounds
2.1%
amino acids, glucose, vitamins, hormones & lipoproteins .

Inorganic compounds
0.9%
sodium, potassium & calcium salts.

2- Formed Elements

- The formed elements include the blood cells which are
- a- Erythrocytes or Red Blood Corpuscles (R.B.Cs.)
- b-Leukocytes or White Blood Cells (W.B.Cs.)
- c-Platelets.
- These cells are developed from the hematopoietic stem cells in the (Bone marrow & lymphoid tissues).

Formed Elements

Erythrocytes or R BCs

- normal range of RBCs is about (3.9-5.5) million /ml In ♀ and (4.1-6.0) million /ml In ♂. (7-8) μm in diameter

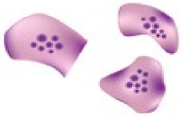
Leukocytes or (W.B.Cs)

normally between 4×(10)9 and 1.1×(10)10 white blood cells in a liter of blood

Plat

Constitute (200,000-400,000) per microliter of blood

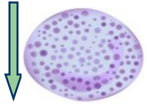
2–3 μm in diameter,
life span about (5-9)days in blood



Granulocytes

Nutrophils
60-70)%
diameter
(10–12) μm ,6-7hrs

Eosinophils
(2-4)%
diameter
(10–12) μm



Basophils
less than(1)%
diameter
(12–15) μm

Agranulocytes

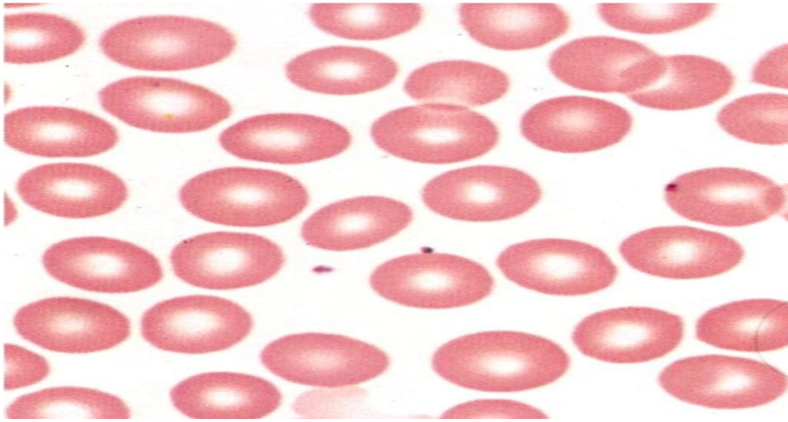
Monocytes

(3-8)%
diameter
(12–20) μm

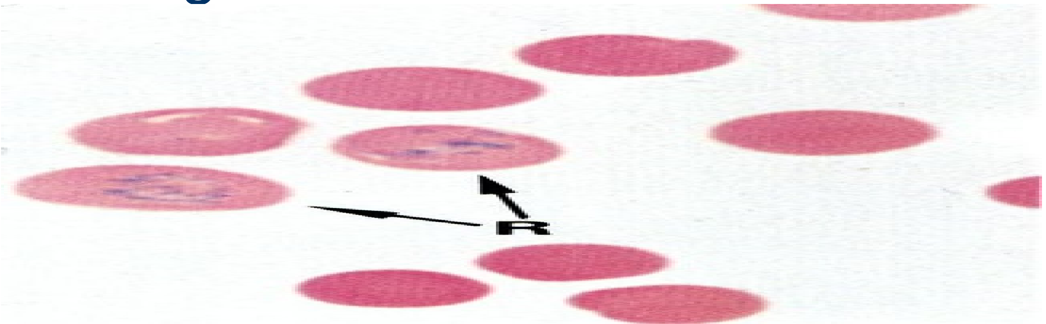
Lymphocytes
Diameter (7-8) μm
a- B cells.
b-T cells.
c-Natural killer cells

a-Erythrocytes or red blood cells (RBC)s:

- Specialized for oxygen & carbon dioxide transport between the lungs and other tissues. RBCs are anucleate cells, packed with oxygen- carrying protein (hemoglobin). The normal range of RBCs is about (3.9-5.5) million /ml In ♀ and about (4.1-6.0) million /ml In ♂.



- RBCs precursors have nuclei and cell organelles but gradually they lose them in mature RBCs, the stage before maturation RBCs is called as reticulocytes it still has the remnants of ribosomal RNA, so contains basophilic precipitations in it's cytoplasm when the cell lose these remnants it become mature RBC and released to blood stream that form about (1%) of the total number of circulating RBCs.

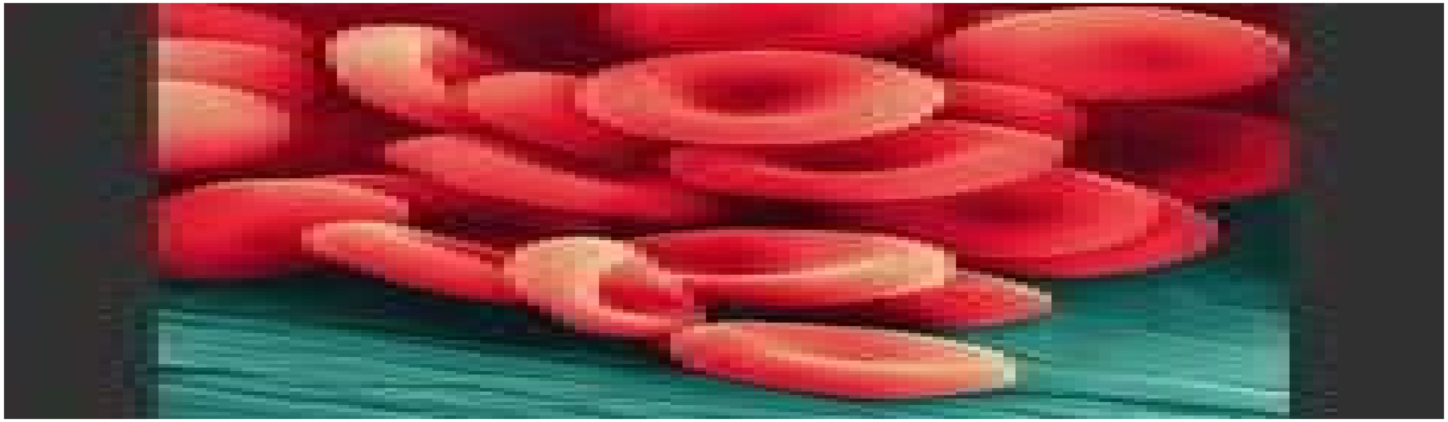


RBCs are characterized by:

- 1- The cells are biconcave disk shaped. This biconcave shape provides the erythrocytes a large surface to volume ratio, thus facilitating gas exchange.

- 2-The cytoplasm packed with (hemoglobin).
- 3- Have no nuclei and organelles.

- 4- Human erythrocytes are arranged about (7-8) µm in diameter.(If the diameter greater than 9 µm is called macrocyte, while those less than 6 Mm is called microcyte).
- 5-Their life span is (120) days.



2- Leukocytes or WBCs

- These cells differ from the erythrocytes in that:
- 1- the cells contain nuclei and organelles.
- 2- Can move by amoeboid movement, and are involved in the defense function against foreign materials. Their number varies according to age, sex and physiological conditions.

- There are normally between $4 \times (10)9$ and $1.1 \times (10)10$ white blood cells in a liter of blood, they migrate to the tissues, where they perform multiple functions.
- - These cells can be divided according to the presence and the type of granules in their cytoplasm & the shape of the nuclei in to TWO groups of cells :
- 1-Granulocytes.
- 2-Agranulocytes.

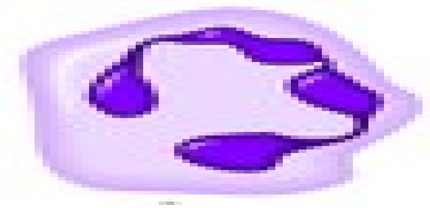
1- Granulocytes

They have nuclei of (2 or more) lobes & the cytoplasm contain 2 types of granules are :

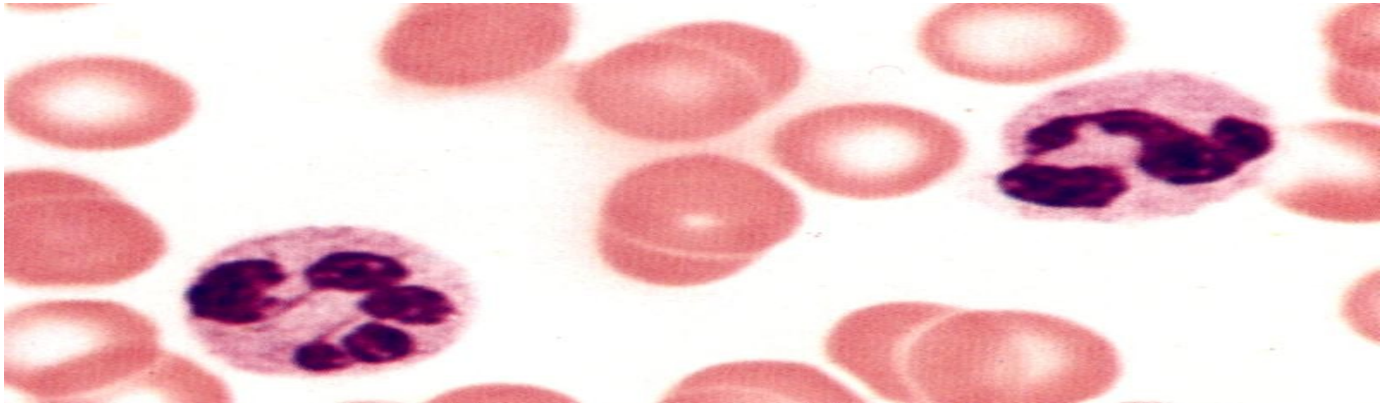
- Specific granules these granules are membrane-bound enzymes.
- Azurophilic granules (are lysosomes)
- types of granulocytes:

- 1-Neutrophils.
- 2-Eosinophils.
- 3-Basophils.

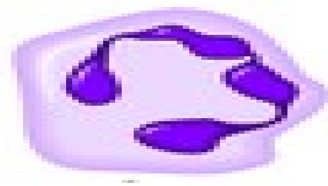
1-Nutrophils:



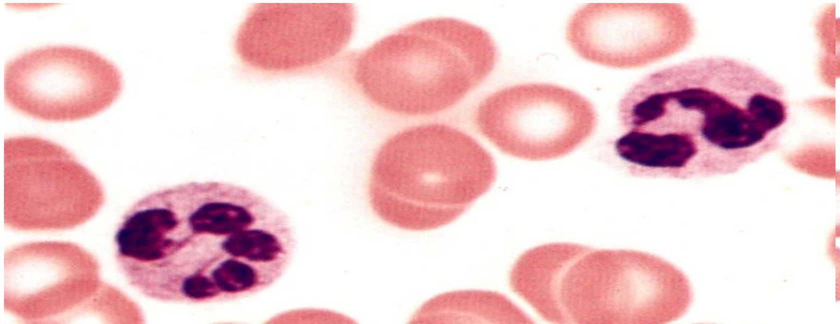
- make about(60-70)% of circulating leukocytes & have diameter about (10–12) μm and characterized by the following:-
- 1-the cells have nuclei consist of (2-5) lobes.

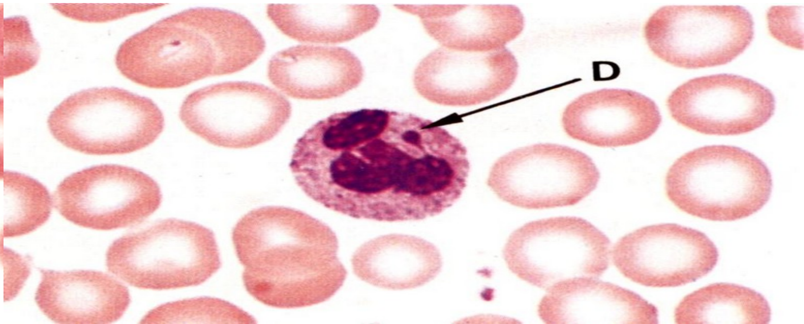


1-Nutrophils:

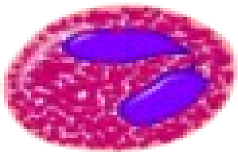


- 2-the cytoplasm appear pink in color and contains 2 types of granules are:
- a-specific neutrophilic granules (small granules).
- b-azurophilic granules are primary lysosomes.
- The cells have life span (6-7) hours in blood .
- - these cells are the first line of cellular defense against bacterial infection.

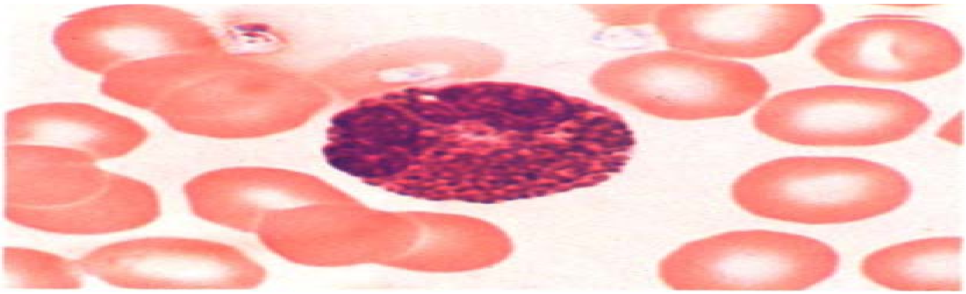




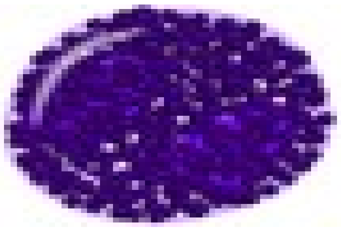
2-Eosinophils



- constitute about (2-4)% of blood WBCs & are about (10–12) μm in diameter.
- 1-have bilobed nuclei.
- 2-the cytoplasm contains large specific granules & similar in size that stain (orange) also contains azurophilic granules, so the cytoplasm appear red in color.
- -these cells play important role in allergic reactions.

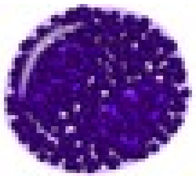


3-Basophils

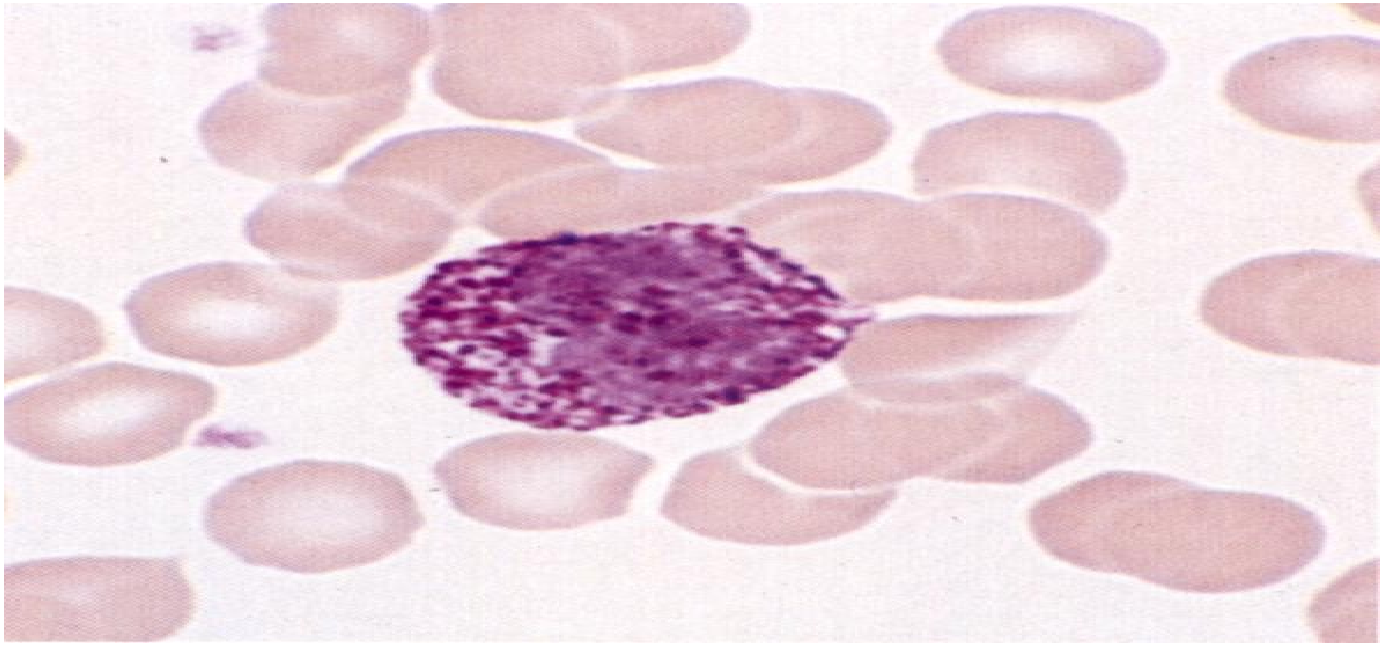


- constitute less than(1)% of blood WBCs & about (12-15) µm in diameter.
- 1-contain irregular lobes or S shaped nuclei.
- 2-the cytoplasm contains variable sized violet to black specific granules contain HEPARIN & HISTAMIN also contain few azurophilic granules, so the cytoplasm appear BLUE in color.

3-Basophils

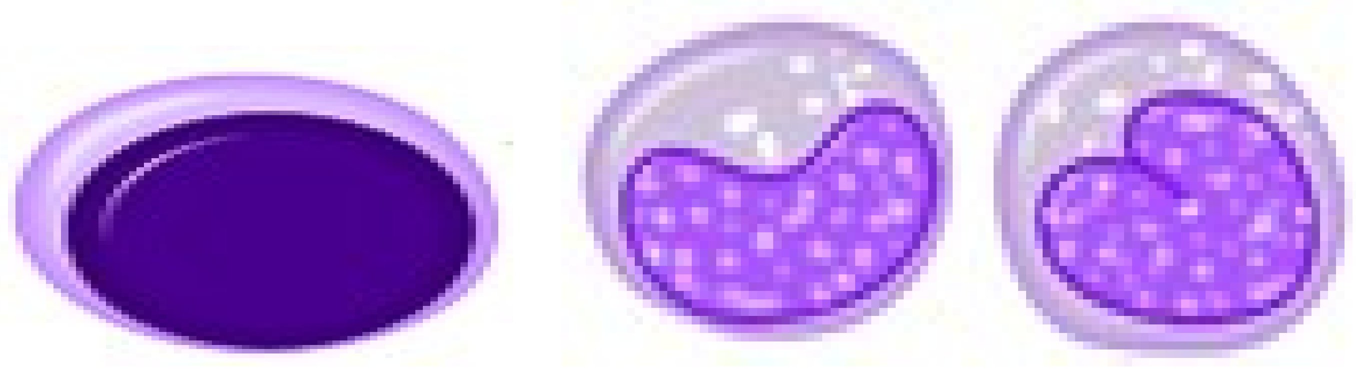


- These cells found at the site of inflammation & release anticoagulant heparin.



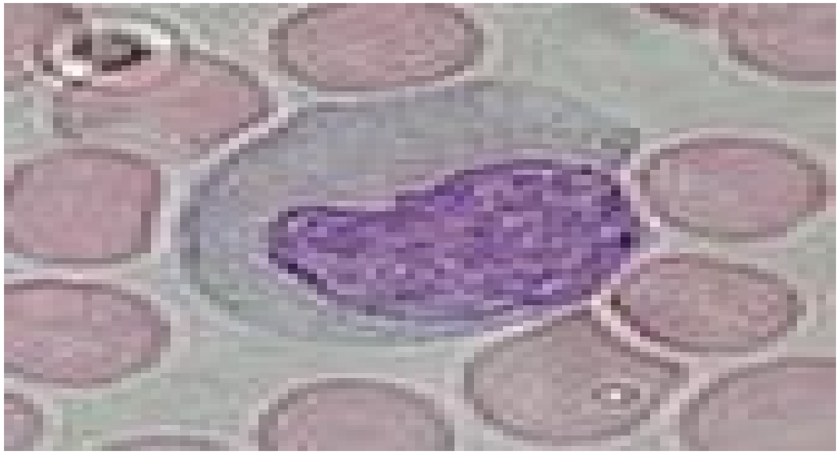
2-Agranulocytes

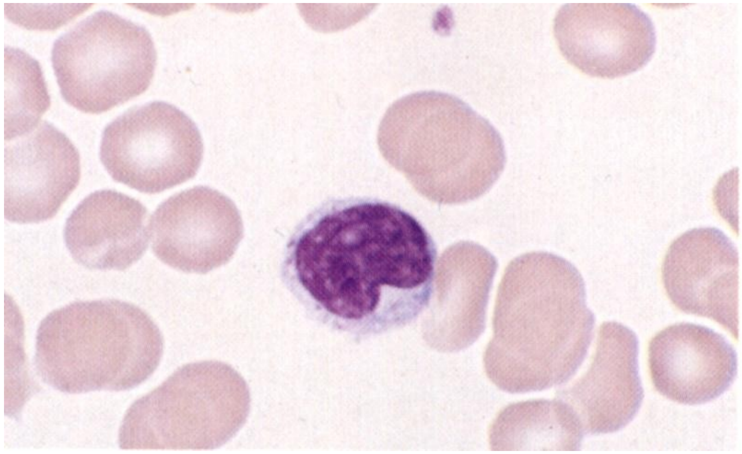
- Do not have specific granules but have more azurophilic granules & the nuclei are rounded or indented.
- They are of 2 types:
- 1-Monocytes.
- 2-Lymphocytes.



1-Monocytes

- 1-constitute (3-8)% of circulating leukocytes and have diameter about (12-20) µm.
- 2-have kidney shaped nuclei, located peripherally .
- 3-have basophilic cytoplasm.

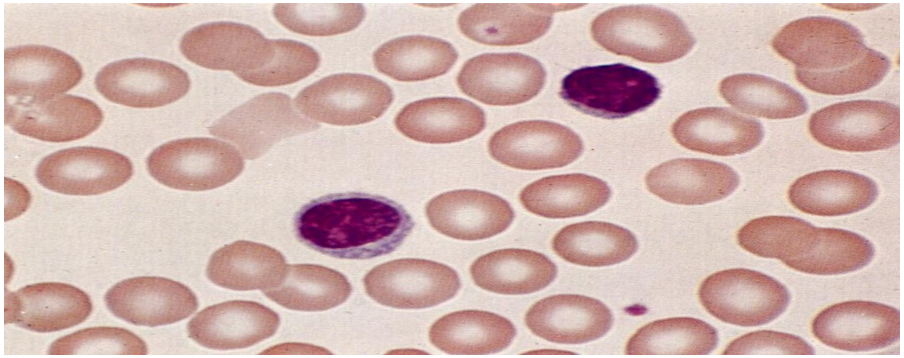




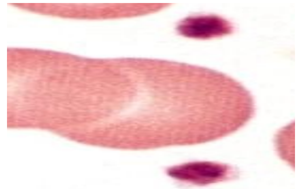
2-Lymphocytes

1- Lymphocytes are much more common in the lymphatic system.

- 2-have diameter about (7-8) µm.
- 3- Lymphocytes are distinguished by having a large deeply staining nucleus which may be eccentric in location, and a relatively small amount of cytoplasm.
- 4- in general there are three types of lymphocytes
- a- B cells.
- b-T cells.
- c-Natural killer cells.



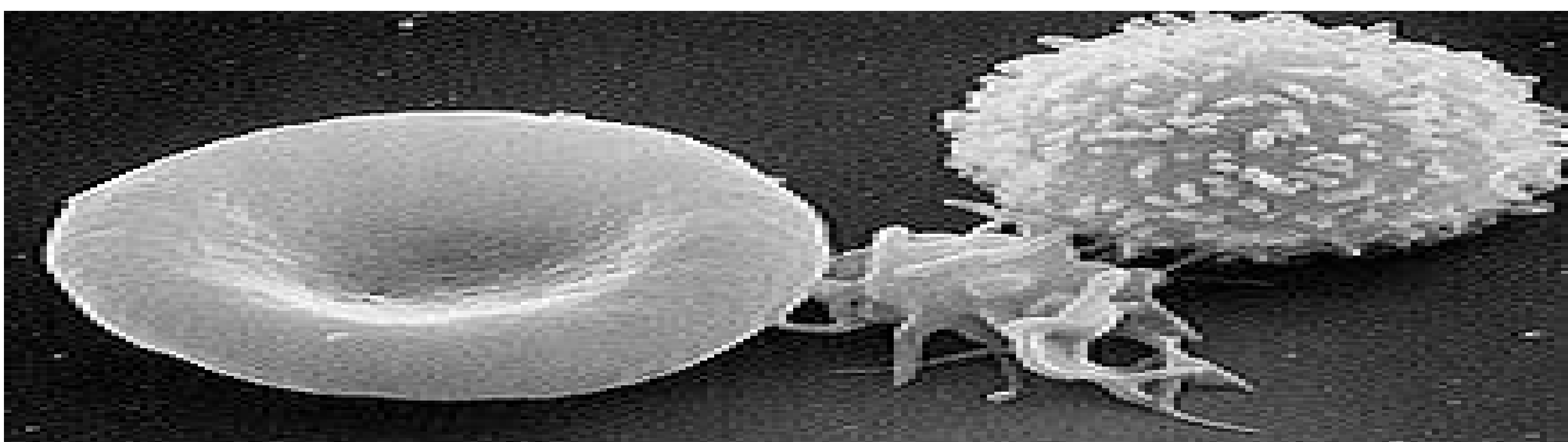
3-Platelates



1- Are small, regularly shaped clear non nucleated disk-like cell fragments, constitute (200,000-400,000) per microliter of blood

2- they are about 2–3 µm in diameter,

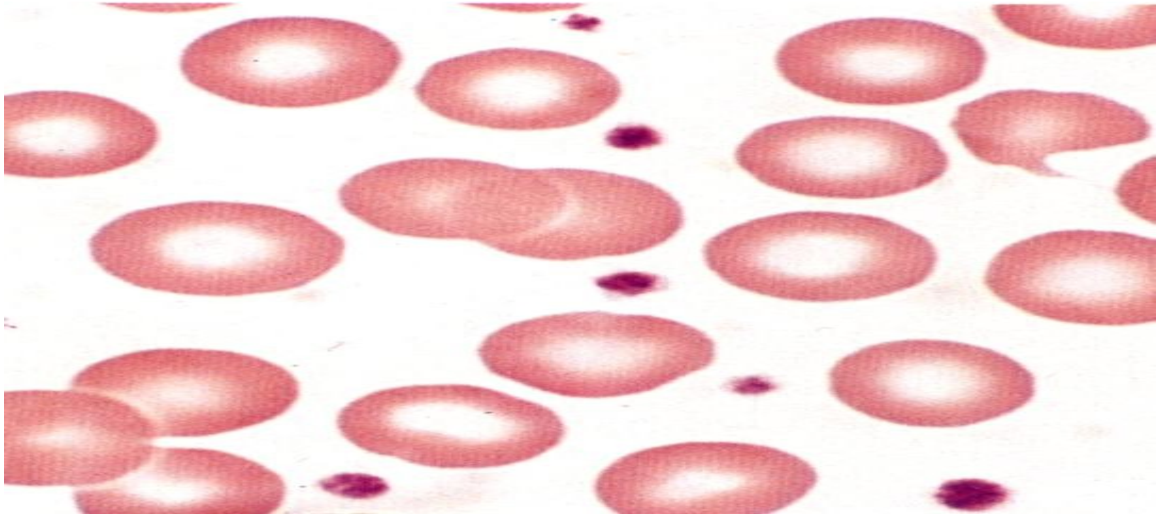
3- have a life span about (5-9)days in blood.



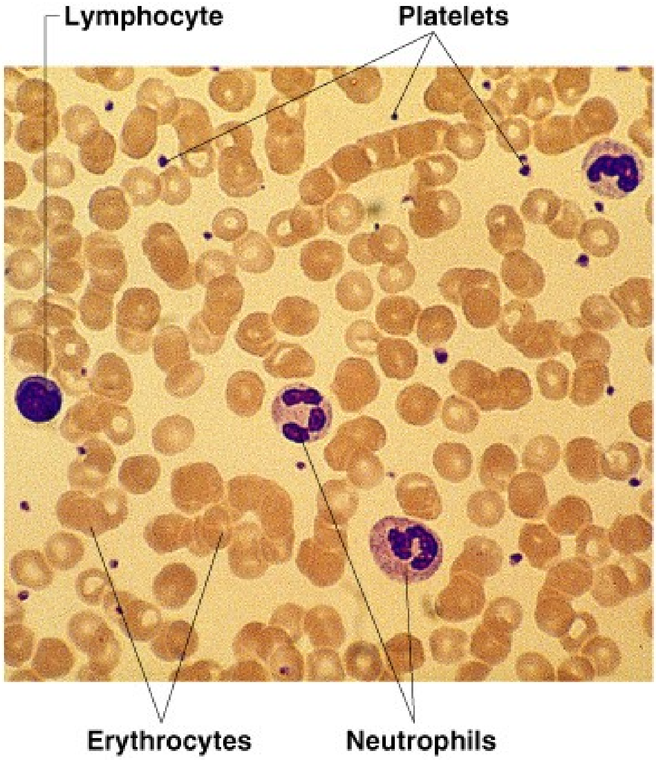
3-Platelates

4- They appear in clumps, contain granules & mitochondria.

5-The main functions of platelates is controlling hemorrhage & play an important role in blood clotting.



Blood smear



Thanks